

ESSA ALBUQUERQUE

Lead Software Developer | Manufacturing Solutions | Business Process Automation | CAD/CAM Expert

Dubai, UAE | +971 56 935 9470 | INDIA +91 771 007 5869 (WhatsApp)

essafarhaan@yahoo.com | <https://essafarhaan.github.io/Portfolio/> |

<https://www.linkedin.com/in/essaalbuquerque>

Lead Software Developer with 29 years of experience designing, developing, and delivering enterprise software solutions for manufacturing, engineering, construction, and industrial organizations.

Specialized in business process automation, manufacturing systems, engineering workflow digitization, CAD/CAM integration, and enterprise application development.

Proven track record of independently architecting and delivering mission-critical platforms including AutoDoor and Dorium, transforming complex engineering and manufacturing processes into scalable digital solutions. Extensive expertise in ASP.NET Core, SQL Server, React, Angular, cloud technologies, enterprise integrations, and digital transformation initiatives.

Recognized for bridging the gap between commercial, engineering, and manufacturing operations through software innovation and automation.

Core Competencies & Technologies

Manufacturing Systems • Enterprise Applications • Business Process Automation • CAD/CAM Integration • Engineering Automation • Solution Architecture • Technical Leadership • Digital Transformation • ASP.NET Core MVC • C# • SQL Server • Oracle • React • Angular • JavaScript • jQuery • REST APIs • AutoCAD • SAP ABAP • Azure • AWS • Git • Agile Development

Work Experience

Bin Ghurair Metal Industries Ltd | UAE

Dec 2025 - Present

Lead Software Developer – Manufacturing Solutions

- Architecting and developing **Dorium**, an enterprise platform for the hollow metal door and frame manufacturing industry.
- Designing integrated workflows covering inquiry management, estimation, quotations, engineering coordination, and shop drawing generation.
- Driving digital transformation by replacing manual and fragmented processes with a centralized ASP.NET Core and SQL Server platform.
- Reduced engineering turnaround from days to hours by enabling approximately 30% of project data to be captured during sales and estimation.
- Developing engineering automation solutions that generate shop drawings and manufacturing documentation directly from business data.

Al Osool Bldg. Metallic Const. Ind. LLC | UAE

May 2025 – Present

Senior Software Engineer

- Assessed and modernized legacy manufacturing applications supporting business and operational workflows.
- Designed a unified ASP.NET Core platform to streamline estimation, engineering, and manufacturing processes.
- Led digital transformation initiatives focused on workflow automation and system modernization

Sirte Oil Company | Libya

May 2017 – Oct 2024

Senior Web Developer

- Designed, developed, and maintained enterprise web applications supporting multiple business departments across the organization.
- Developed a company-wide Corporate Directory System used by employees to access interdepartmental contact information and improve internal communication.
- Architected and implemented a Timesheet Management System to support workforce time tracking, reporting, and operational oversight across departments.
- Developed a Training & Development Management System to manage mandatory training programs, certification records, compliance requirements, and annual training budgets.
- Built an HSE Management System for the Fire & Safety Department to support daily inspections, compliance activities, reporting, and operational monitoring.
- Collaborated with stakeholders across multiple departments to analyze business requirements and deliver software solutions that improved efficiency, reporting, and process automation.

H & I Infotech Co. | India

Software Development Manager | India

Jan 2014 - Apr 2017

- Led software development teams responsible for the successful delivery of enterprise applications across multiple projects.
- Managed project planning, resource allocation, technology evaluation, and development activities while ensuring alignment with business objectives.
- Mentored development teams and promoted best practices in software engineering, Agile methodologies, and solution delivery.

Mammut Building Systems | UAE

Jul 2006 - Oct 2013

Team Leader

- Led the design and development of custom software solutions supporting pre-engineered steel building operations.
- Architected and developed a Project Information Form (PIF) system that captured and managed complete building requirements through a structured engineering workflow.
- Integrated AutoCAD automation to generate proposal and approval drawings, improving engineering productivity and reducing manual drafting effort.
- Managed software planning, development, testing, deployment, and support activities while leading development initiatives across engineering and business functions.

Al Kuhaimi Group | Saudi Arabia Feb 2000 - Jul 2006

Software Developer

Designed and developed **AutoDoor**, an enterprise manufacturing platform supporting engineering, production, and manufacturing operations for hollow metal door and frame products.

Key Achievements

- Independently architected and developed AutoDoor, a Windows-based manufacturing solution used across engineering and production departments.
- Integrated with PTC Pro/ENGINEER to automate engineering workflows and manufacturing data preparation.
- Developed automated Bill of Materials (BOM) generation capabilities, improving engineering accuracy and reducing manual effort.
- Created manufacturing order generation processes complete with engineering drawings and production documentation.
- Automated engineering calculations and manufacturing workflows supporting door and frame production operations.
- Generated CNC-ready production data and DXF outputs for punching and bending operations.
- Eliminated dependence on repetitive manual engineering tasks through software-driven automation.
- Supported approximately 14+ engineering and production users while serving as the sole architect, developer, and maintainer of the platform.

Technologies: Visual Basic, XML, AutoCAD, PTC Pro/ENGINEER API, CAD/CAM Integration, DXF Generation, Manufacturing Automation

Software Developer

Jan 1997 - Feb 2000

Zamil Steel Buildings | Saudi Arabia

- Led the development of a new application system, overseeing project requirements, resource allocation, and task assignments to ensure timely delivery.
- Directed system integrations and application updates, enhancing process reliability and reducing system errors across key operational areas.

Core Skills

C, C++, Java, C#, .NET 6.0/7.0/8.0, VB.NET, ASP.NET Core MVC, SAP ABAP, jQuery, JavaScript, Bootstrap, UI/UX Design, Website & Digital Design, Branding & Visual Design, JSON, jQuery, ReactJS, AngularJS, HTML, DHTML, CSS, XML, XAML, SQL Server, Oracle, SAP Hana, MongoDB, T-SQL, PL/SQL, GIT, GitHub, TFS, Azure, AWS, Google Cloud, Agile, Scrum and Waterfall

Education

Microsoft .NET Technologies Certification
Learning Solutions, Dubai

Diploma in CAD/CAM Technologies
Sonata, Bangalore

Diploma in Computer Applications & Programming
Computer World, Mumbai

Diploma in Mechanical Drafting
Fr. Agnel Polytechnic, Mumbai